

**FMUSER FUQ-4653U 4K Encoder Modulator Dual HDMI 2.0 2xASI
32xIP Inputs DVB-C-T ASI-IP Outputs H.265 HEVC Encoding IPTV
Broadcast Headend**



Up to 2160P 60Hz RGB/YCbCr 4:4:4
H.265 HEVC/H.264 AVC
Video Resolution Upscale Transform
HDCP 2.2 and HDCP 1.4
HDMI 2.0

Logo/Caption/QR Code Insertion



Product Overview

The FMUSER FUQ-4653U is a professional 4K encoder modulator supporting dual HDMI 2.0, 2xASI, and 32xIP inputs with DVB-C/T RF, ASI, and IP outputs. It performs H.265/HEVC encoding and DVB modulation in a 1U rack-mountable chassis. The device features web-based NMS for remote control and supports PSI/SI

table generation. Ideal for digital terrestrial television (DTT) headends and IPTV distribution systems requiring 4K UHD processing.

Key feature

Flexible Audio: MPEG-1 Layer 2 and LC-AAC audio encoding options

Specifications

Input	1×HDMI input (2.0) with 2 HDMI port, HDCP 2.2/1.4	1×HDMI input (2.0) with 2 HDMI port, HDCP 2.2/1.4	1×HDMI input (2.0) with 2 HDMI port, HDCP 2.2/1.4	1×HDMI input (2.0) with 2 HDMI port, HDCP 2.2/1.4	1×HDMI input (2.0) with 2 HDMI port, HDCP 2.2/1.4	1×HDMI input (2.0) with 2 HDMI port, HDCP 2.2/1.4
Input	2 ASI input, BNC interface	2 ASI input, BNC interface	2 ASI input, BNC interface	2 ASI input, BNC interface	2 ASI input, BNC interface	2 ASI input, BNC interface
Input	32 IP input over UDP and RTP protocol	32 IP input over UDP and RTP protocol	32 IP input over UDP and RTP protocol	32 IP input over UDP and RTP protocol	32 IP input over UDP and RTP protocol	32 IP input over UDP and RTP protocol
Video Input	Resolution	Resolution	Resolution	Resolution	Chroma	Chroma
Video Input	3840×2160_60P, 3840×2160_59.94P, 3840×2160_50P, 3840×2160_30P, 3840×2160_29.96P, 3840×2160_25P, 3840×2160_24P, 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P, 1920×1080_30P, 1920×1080_25P, 1920×1080_24P	3840×2160_60P, 3840×2160_59.94P, 3840×2160_50P, 3840×2160_30P, 3840×2160_29.96P, 3840×2160_25P, 3840×2160_24P, 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P, 1920×1080_30P, 1920×1080_25P, 1920×1080_24P	3840×2160_60P, 3840×2160_59.94P, 3840×2160_50P, 3840×2160_30P, 3840×2160_29.96P, 3840×2160_25P, 3840×2160_24P, 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P, 1920×1080_30P, 1920×1080_25P, 1920×1080_24P	3840×2160_60P, 3840×2160_59.94P, 3840×2160_50P, 3840×2160_30P, 3840×2160_29.96P, 3840×2160_25P, 3840×2160_24P, 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P, 1920×1080_30P, 1920×1080_25P, 1920×1080_24P	RGB, YCbCr 4:4:4, YCbCr 4:2:2, YCbCr 4:2:0 (2160P_50/60Hz only)	RGB, YCbCr 4:4:4, YCbCr 4:2:2, YCbCr 4:2:0 (2160P_50/60Hz only)

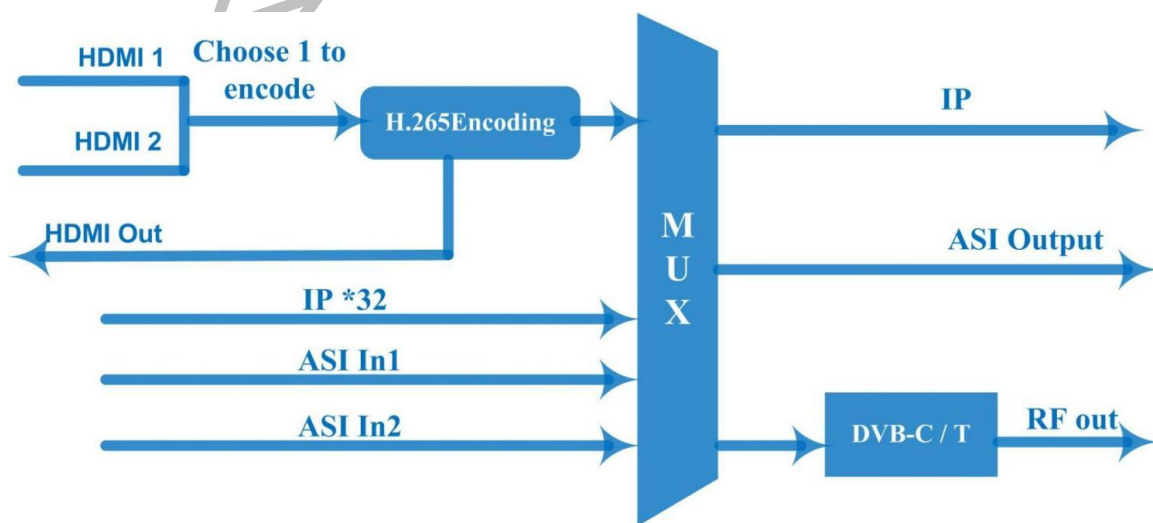
Video Encoding	Encoding	Encoding	H.265/HEVC, H.264/AVC	H.265/HEVC, H.264/AVC	H.265/HEVC, H.264/AVC	H.265/HEVC, H.264/AVC
Video Encoding	Chroma	Chroma	4:2:2, 4:2:0	4:2:2, 4:2:0	4:2:2, 4:2:0	4:2:2, 4:2:0
Video Encoding	Bit-rate	Bit-rate	1Mbps~23Mbps	1Mbps~23Mbps	1Mbps~23Mbps	1Mbps~23Mbps
Video Encoding	Rate Control	Rate Control	CBR	CBR	CBR	CBR
Audio Encoding	Encoding	Encoding	MPEG-1 Layer 2, LC-AAC	MPEG-1 Layer 2, LC-AAC	MPEG-1 Layer 2, LC-AAC	MPEG-1 Layer 2, LC-AAC
Audio Encoding	Sampling rate	Sampling rate	48KHz	48KHz	48KHz	48KHz
Audio Encoding	Bit-rate	Bit-rate	64Kbps~256Kbps each channel	64Kbps~256Kbps each channel	64Kbps~256Kbps each channel	64Kbps~256Kbps each channel
Multiplexing	2 ASI input multiplexed with local 1 channel of TS	2 ASI input multiplexed with local 1 channel of TS	2 ASI input multiplexed with local 1 channel of TS	2 ASI input multiplexed with local 1 channel of TS	2 ASI input multiplexed with local 1 channel of TS	2 ASI input multiplexed with local 1 channel of TS
Multiplexing	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically
Modulator Section	DVB-C	Standard	Standard	J.83A, J.83B	J.83A, J.83B	J.83A, J.83B
Modulator Section	DVB-C	MER	MER	≥43dB	≥43dB	≥43dB

Modulator Section	DVB-C	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulator Section	DVB-C	RF output level	RF output level	-25.0~ -1dbm, 0.1db step	-25.0~ -1dbm, 0.1db step	-25.0~ -1dbm, 0.1db step
Modulator Section	DVB-C	Symbol rate	Symbol rate	5.0 - 7.0 Msps	5.0 - 7.0 Msps	5.0 - 7.0 Msps
Modulator Section	DVB-C			J.83A	J.83A	J.83B
Modulator Section	DVB-C	Constellation	Constellation	16/32/64/128/ 256 QAM	16/32/64/128/ 256 QAM	64/ 256 QAM
Modulator Section	DVB-C	Bandwidth	Bandwidth	8M	8M	6M
Modulator Section	DVB-C	RF out	RF out	1*RF DVB-C out	1*RF DVB-C out	1*RF DVB-C out
Modulator Section	DVB-T	Standard	Standard	EN300744	EN300744	EN300744
Modulator Section	DVB-T	FFT mode	FFT mode	2K/4K/8K	2K/4K/8K	2K/4K/8K
Modulator Section	DVB-T	Bandwidth	Bandwidth	6M, 7M, 8M	6M, 7M, 8M	6M, 7M, 8M
Modulator Section	DVB-T	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulator Section	DVB-T	Guard Interval	Guard Interval	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32
Modulator Section	DVB-T	Code rate	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulator Section	DVB-T	MER	MER	≥42dB	≥42dB	≥42dB
Modulator Section	DVB-T	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulator Section	DVB-T	RF output level	RF output level	-28~-3dbm, 0.1db step	-28~-3dbm, 0.1db step	-28~-3dbm, 0.1db step

Modulator Section	DVB-T	RF out	RF out	1*RF DVB-T out	1*RF DVB-T out	1*RF DVB-T out
Stream output	1*ASI output through 4 BNC interfaces	1*ASI output through 4 BNC interfaces	1*ASI output through 4 BNC interfaces	1*ASI output through 4 BNC interfaces	1*ASI output through 4 BNC interfaces	1*ASI output through 4 BNC interfaces
Stream output	1 MPTS over UDP/RTP, 1000M/100 M Base-T Ethernet interface (unicast/multicast)	1 MPTS over UDP/RTP, 1000M/100 M Base-T Ethernet interface (unicast/multicast)	1 MPTS over UDP/RTP, 1000M/100 M Base-T Ethernet interface (unicast/multicast)	1 MPTS over UDP/RTP, 1000M/100 M Base-T Ethernet interface (unicast/multicast)	1 MPTS over UDP/RTP, 1000M/100 M Base-T Ethernet interface (unicast/multicast)	1 MPTS over UDP/RTP, 1000M/100 M Base-T Ethernet interface (unicast/multicast)
System function	2 versions of front panel: Option1: with LCD and Keyboard; Option2 without LCD or Keyboard.	2 versions of front panel: Option1: with LCD and Keyboard; Option2 without LCD or Keyboard.	2 versions of front panel: Option1: with LCD and Keyboard; Option2 without LCD or Keyboard.	2 versions of front panel: Option1: with LCD and Keyboard; Option2 without LCD or Keyboard.	2 versions of front panel: Option1: with LCD and Keyboard; Option2 without LCD or Keyboard.	2 versions of front panel: Option1: with LCD and Keyboard; Option2 without LCD or Keyboard.
System function	LCD/keyboard operating(optional), web NMS supporting	LCD/keyboard operating(optional), web NMS supporting	LCD/keyboard operating(optional), web NMS supporting	LCD/keyboard operating(optional), web NMS supporting	LCD/keyboard operating(optional), web NMS supporting	LCD/keyboard operating(optional), web NMS supporting
System function	Ethernet software & hardware upgrade	Ethernet software & hardware upgrade	Ethernet software & hardware upgrade	Ethernet software & hardware upgrade	Ethernet software & hardware upgrade	Ethernet software & hardware upgrade
Miscellaneous	Dimension (W× L× H)	Dimension (W× L× H)	Dimension (W× L× H)	482mm×455mm×44mm	482mm×455mm×44mm	482mm×455mm×44mm
Miscellaneous	Approx weight	Approx weight	Approx weight	4kg	4kg	4kg

Miscellaneous	Temperature	Temperature	Temperature	0~45℃(work), -20~80℃ ■Storage ■	0~45℃(work), -20~80℃ ■Storage ■	0~45℃(work), -20~80℃ ■Storage ■
Miscellaneous	Power	Power	Power	AC 100V-220V±10%, 50/60Hz	AC 100V-220V±10%, 50/60Hz	AC 100V-220V±10%, 50/60Hz
Miscellaneous	Consumption	Consumption	Consumption	25W	25W	25W

Principle Chart



Resolution Upscale



About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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